EPSR-test-ex76 sdfcube DMSO-water (distinct), nsumt = 5 without a cube file

Program(s): sdfcube | chained passes: 1 | verification: ALL IDENTICAL (1 pass)

What this case tests

Exercises `sdfcube` [1] for two different molecule types (spatial density of water around DMSO) and the start of an accumulation when no previous cube file exists.

Code exercised

Fortran (original, fortran_src/):

- `sdfcube.f`: `run_sdfcube`, `sdfcube` (grid histogram), `initcube`, `outcube` (cube file, accumulation)
- `sharm_routines_rev1.f`: `form_eulerangles`; `auxiliary_routines.f`: `rotateeuler`; `process_symbols.f`: `check_symbol`
- `setup_symbols_generic.f`: `setup_symbols_sdfcube`

Rust (port, rust/epsr-core/src/):

- `tools/sdfcube.rs`: `run`, `sdfcube`, `initcube`, `outcube`
- `tools/sharm_routines.rs`: `form_eulerangles`, `rotateeuler`, `write_mol_xyz`; `tools/sharm.rs`: `read_axis`, `average_molecule`
- `symtab_tools.rs`: `SDFCUBE` keyword table

Physical system

The EPSR-test configuration `200DMSO400Water.ato`: 200 DMSO (atoms S, C, C, O, 6 H) and 400 water molecules (O1, H1, H1; SPC/E parameters [2]), 3200 atoms at 300 K in a cubic box of side 33.12 Å, a model of the DMSO-water mixtures of Soper and Luzar [3]. `sdfcube` histograms the centres of the second molecules, rotated into the frame of each central molecule (Euler angles from the axis definitions), on a cubic grid and writes the spatial density function as a Gaussian `.cube` file together with the averaged molecules (`-first.xyz`, `-second.xyz`). Here water centres (O1) around DMSO (centre = mass centre of S, O and C; z axis towards O, x axis towards the C-C midpoint) on a 21×21×21 grid (`mpixel 10`, `rmax 8`), with the "molecule 1 at origin" pass of distinct types.

Inputs

Input files written by hand from the keyword tables (`setup_symbols_*`), one `keyword value` per line, ended by `q`; `200DMSO400Water.ato` copied from EPSR-test. `dmsowat.SDFCUBE.dat` with `nsumt 5` and `averagemol n`; compared with ex75 it adds distinct molecule types and a three-type central group.

How the accuracy was verified

`commands.txt` (1 program call, run once) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; no masking except where noted below). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in 1 pass.

One pass. With `nsumt 5` but no cube file, `initcube` resets `nsumt` to 0, so the cube is a fresh accumulation and the saved input file has `nsumt 1` (reference `dmsowat.SDFCUBE.dat`). Compared: `dmsowat.SDFCUBE.cube`, `-first.xyz`, `-second.xyz` and the updated input file.

References

[1] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).

[2] H. J. C. Berendsen, J. R. Grigera and T. P. Straatsma, "The missing term in effective pair potentials", J. Phys. Chem. 91, 6269-6271 (1987).

[3] A. K. Soper and A. Luzar, "A neutron diffraction study of dimethyl sulphoxide-water mixtures", J. Chem. Phys. 97, 1320-1331 (1992).

[4] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.